MIL-T-43187B(GL)
15 February 1979
SUPERSEDING
MIL-T-43187A(GL)
11 December 1967

MILITARY SPECIFICATION

TESTER, LEAK, INSULATED BOOT

This specification is approved for use by the Natick Research and Development Command, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of insulated boot leak tester (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-P-291	-	Paperboard, Wrapping, Cushioning
PPP-T-60	-	Tape: Packaging, Waterproof

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3520

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:52Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

DRAWING

US ARMY NATICK RESEARCH AND DEVELOPMENT COMMAND

2-1-525 - Tester, Leak, Insulated Boot; Assembly Complete

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Alternate components. Components offered as equivalent to components specified hereinafter and on the reference drawings as specific manufacturer's part number or equal shall be functionally equal, or equal or better quality, with the manufacturer's part number identified. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other specified component, and shall not reduce ease of maintenance to it or any other components. Prior to manufacture of the first article, or if none is required, prior to commencing production, the contractor shall submit for the contracting officer's approval a list identifying each proposed alternate component, together with proof that each listed component complies with requirements specified herein. The contracting officer, at his own option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of the first article or production manufacture as applicable, and does not relieve the contractor of responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.2 First article. When specified (see 6.2), the contractor shall furnish a sample for first article inspection and approval (see 4.2 and 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:52Z
Check the source to verify that this is the current version before use.

3.3 Materials and components. The materials and components shall be as specified herein and shown on the applicable drawing. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided the completed item complies with all of the provisions of this specification (see 6.4).

3.3.1 Conduit, fittings, and electric wire. Electrical conduit shall be rigid or flexible steel, or electrical metallic tube in accordance with current commercial practice. All electrical wire shall be commercial heat resistant grade, thermoplastic insulated. External conduit fittings shall be a waterproof type.

3.4 Design and construction. The design and construction of the tester shall be as specified herein and as shown on the drawing. Unless otherwise specified (see 6.2) the tester shall operate on a nominal 120 volts, 60 hertz, single phase. The compressor shall be direct motor driven with a totally enclosed motor. The air system and water tank shall not leak, and there shall be no evidence of failure of the compressor, switch, pressure accuated motor control, gages, stopcocks and boot valve when the tester is tested as specified in 4.5.1.

3.4.1 Air distribution assembly. Each air distribution assembly shall have no visible evidence of leakage when tested as specified in 4.5.2.

3.5 Contract data requirements. The contract data requirements shall be as specified (see 6.2 and 6.2.1).

3.6 Marking for identification. The units shall be marked in accordance with the manufacturer's commercial practice and shall include manufacturer's name, date of manufacture, and electrical characteristics.

3.7 Workmanship. The units shall be free from defects such as fractures, splits, punctures, tears, dents, creases, deteriorations or malformation. There shall be no sharp edges, slivers, burns or projections. Threaded fasteners shall not be broken, stripped, fractured or loose. Rivets shall completely fill the holes, with full size heads neatly finished and in full contact with the surface of the riveted members and shall not be loose.

3.7.1 Welding. The surface of parts to be welded shall be free from oxide, paint, grease, and other foreign matter. Welds shall be continuous, sound, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux shall be removed from the finished welds. Soldering shall be complete and adherent, with all flux and residual removed and shall contain no pinholes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:52Z
Check the source to verify that this is the current version before use.

3.7.2 Electrical wiring. Wiring shall not be cut, abraded, or have insulation stripped, and shall be joined at the terminals. Wiring shall have adequate slack to provide strain relief.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3)
2. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article is required (see 3.2), the unit shall be examined as specified in 4.4.3.1 and 4.4.3.2 and tested as specified in 4.5.1 and 4.5.2. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. The contractor shall insure that all components and materials used are manufactured, examined, and tested in accordance with referenced specifications, drawings, and standards unless excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection and testing.

4.4.2.1 Visual examination of operations. In-process examination shall be made of the surface preparation prior to welding for compliance with 3.7.1. Whenever non-conformance is noted, correction shall be made to the applicable process and all items processed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:52Z
Check the source to verify that this is the current version before use.

4.4.2.2 Air distribution assembly testing. Prior to the attachment of hose and rubber tubing, each air distribution assembly shall be tested to determine compliance with 3.4.1. Any evidence of leaks shall be cause for rejection of the assembly.

4.4.3 End item inspection. The lot shall be all the units submitted for inspection at one time. The sample unit for examination shall be one completely fabricated leak tester.

4.4.3.1 Visual examination. Examination of the end item shall be in accordance with defects listed in table I. The inspection level shall be II with an acceptable quality level (AQL) of 4.0 for major defects and 10.0 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not clean, smooth		X
Design, construction and workmanship, general (applicable to all components assemblies)	Any part missing or not specified type	X	
	Split, punctured, torn, dented, creased, deteriorated or malformed	X	
Welding and brazing (where required)	Missing, incomplete or not adherent	X	
	Not clean (flux or residue not removed) or contains pinholes		X
Soldering (when applicable)	Missing, incomplete or not adherent	X	
	Not clean (flux or residue not removed) or contains pinholes		X
Bolts, nuts, screws, studs and other threaded fasteners	Missing, broken, stripped or fractured	X	
	Threaded fasteners loose		X
Rivets (where applicable)	Missing, broken, stripped or fractured	X	
	Rivets loose		X
Electrical apparatus wiring	Wiring cut, abraded, insulation stripped, not properly joined or loose at terminal	X	
	Not inclosed in specified conduit where required	X	
	Adequate slack not provided for wiring to relieve strain or excessive insulation stripped from wiring		X

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:52Z
Check the source to verify that this is the current version before use.

TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Marking for identification	Missing, incomplete, not legible	X	

4.4.3.2 Dimensional examination. Inspection shall be made of the end item for compliance with the dimensions specified. The inspection level shall be S-2 with an AQL of 6.5, expressed in terms of defects per hundred units. Any non-conformance shall be classified a defect.

4.4.3.3 Testing of the end item. Each end item shall be tested as specified in 4.5.1 and 4.5.2. Failure to pass either test shall be cause to reject the effected end item.

4.4.4 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking as required by section 5 are complied with. Defects shall be as indicated in table II. The sample unit shall be one shipping container fully packaged, except that it need not be closed. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

TABLE II. Packaging inspection

Examine	Defect
Markings	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Components missing, damaged, or otherwise defective
Workmanship	Inadequate application of components such as blocking and cushioning inadequate, improper, or missing

4.5 Tests.

4.5.1 Operational test. The tester shall be operated for two hours with the water tank filled and with the pressure actuated motor control set between 20 and 40 pounds per square inch (138 and 276 kPa) gage. Any nonconformance with the requirements of 3.4 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:52Z
Check the source to verify that this is the current version before use.

4.5.2 Air manifold assembly test. The air manifold assembly shall be tested hydrostatically at 100 psi (690 kPa) gage. Any evidence of leakage shall constitute failure of this test (see 3.4.1).

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. The air hose shall be disconnected at the air distribution assembly end and the assembly then lifted out of the tank. The toe holddown tube shall be removed from the clips and the toggle bolts replaced. The cover shall be lifted off the 4 legs and the receiver assembly, with all equipment attached thereon, shall then be detached from the leg assemblies. The drain stopper shall be removed from the drain hole.

5.1.1.2 Prepacking. All openings into the motor, switches, compressor, regulator, open swivels, nipples, and valves shall be sealed with tape conforming to type III or IV of PPP-T-60. The gages, the glass air filter bowl, drain cock, valve boot assembly, and the needles (including spare needles) shall be individually wrapped in paperboard conforming to type III, style 1 or 2 of PPP-P-291 and the wrap secured with tape specified herein. The wrap shall extend a minimum of 2 inches (51 mm) beyond the end of each needle to prevent damage during unit packing and in transit. The electric service cord shall be compactly coiled, wrapped in minimum 30 pound (48.8 g/m^2) basis weight (24 X 36-500) kraft paper, and the wrap secured.

5.1.1.3 Unit packing. The toe holddown tube shall be placed under the heel holddown angle and secured. The inside bottom of each tank shall be lined with paperboard specified in 5.1.1.2. The disassembled components, prepacked as specified, together with the parts and tools bag shall then be unit packed in the most compact manner within the tank. The components shall be cushioned so as to preclude movement or damage while in transit. The tank cover shall be secured in place and sealed around the opening with minimum 2 inch (51 mm) width tape specified in 5.1.1.2.

5.1.2 Commercial. The testers shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:52Z
Check the source to verify that this is the current version before use.

5.2.1 Level A packing. Each complete tester, preserved as specified in 5.1, shall be packed in a shipping container conforming to overseas type of PPP-B-601, or class 2, style 2 or 4 of PPP-B-621, except that the requirements for additional battens or bleats shall not apply. Lumber blocking shall be placed at each end of the shipping container to provide a minimum clearance of 0.5 inch (13 mm) between projections on the tank and the inner surfaces of the container. Closure and strapping shall be in accordance with the appendix of the applicable container specification.

5.2.2 Level B packing. Each complete tester, preserved as specified in 5.1, shall be packed in a shipping container conforming to domestic type, style A or B of PPP-B-601; or class 1, style 2 or 4 of PPP-B-621, except that the requirements for additional battens or cleats shall not apply. Lumber blocking shall be placed at each end of the shipping container to provide a minimum clearance of 0.5 inch (13 mm) between projections on the tank and the inner surfaces of the container.

5.2.3 Commercial packing. Testers, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129 and MIL-STD-1188, as applicable.

6. NOTES

6.1 Intended use. The leak tester covered by this specification is intended to be used to test for leaks in all types and sizes of insulated boots that are worn by members of the Armed Forces.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a first article is required (see 3.2, 4.3 and 6.3).
(c) When electrical characteristics are other than specified (see 3.4).
(d) Contract data requirements (see 3.5).
(e) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

6.2.1 Contract data requirements. Any data items to be delivered under any contract covered by this specification should be specifically called for in the contract in accordance with the applicable regulation of the procuring activity (see 3.5 and 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:52Z
Check the source to verify that this is the current version before use.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of the Armed Forces Procurement Regulation ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification (see 3.3).

6.5 Metric equivalent. Metric equivalents indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in customary units shall be controlling.

6.6 Changes from previous issue. Astericks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:

Army - GL

Preparing activity:

Army - GL

Project No. 3520-A031

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:52Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.
NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-T-43187B(GL) Tester, Leak, Insulated Boot

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:52Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
US ARMY NATICK RESEARCH and DEVELOPMENT COMMAND
NATICK, MASSACHUSETTS 01760

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

DRDNA-ES

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD-314



Commander
US Army Natick Research and Development Command
ATTN: DRDNA-ES
Natick, MA 01760

NATICK OP 1 AUG 78
FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:52Z
Check the source to verify that this is the current version before use.